

# Supplementary Material for DescFuse: Description-Supervised Facet-Aware Evidence Fusion for Automated ICD Coding from Long Clinical Notes

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## ABSTRACT

This supplementary material provides implementation settings, reference-corpus statistics, additional predictive metrics, sensitivity analyses, evidence-support diagnostics, computational measurements, and quantitative representation statistics for DescFuse. All reported experiments use the datasets and evaluation protocols defined in the main manuscript.

## S1. Implementation and evaluation details

The supplementary experiments use the data splits, candidate code sets, and evaluation procedure reported in the main manuscript. MIMIC-III Full contains 47,723/1,631/3,372 training/validation/test notes and 8,922 candidate codes. The MIMIC-III top-50 and rare50 settings contain 8,066/1,573/1,729 and 249/20/142 notes, respectively. The MIMIC-IV ICD-9-50 and ICD-10-50 settings contain 170,664/6,406/12,405 and 104,077/3,805/7,368 notes [1, 2]. We lowercase notes, remove non-alphanumeric symbols, and retain up to 4,000 input tokens by default. Validation data select checkpoints and classification thresholds. Test inference scores the complete candidate set in blocks of 256 codes.

Table S1 lists the settings used for the reported model. Each training batch retains all positive codes and samples the remaining candidates uniformly with importance weighting. Validation and test evaluation use the full label space. The three reported runs use independent initializations under the same configuration.

**Table S1**

Implementation and optimization settings for DescFuse.

| Setting                  | Value              | Setting                                    | Value     |
|--------------------------|--------------------|--------------------------------------------|-----------|
| Base language model      | LLaMA-3.1-8B       | Facet count $K$                            | 6         |
| Word embedding dimension | 100                | Facet-state dimension $d$                  | 512       |
| BiLSTM hidden units      | 256                | Prediction MLP hidden units                | 1,024     |
| BiLSTM dropout           | 0.3                | Classifier batch size                      | 8         |
| LoRA rank / scale        | 2 / 32             | Description code batch                     | 128       |
| LoRA peak learning rate  | $2 \times 10^{-4}$ | Cross-facet learning rate                  | $10^{-4}$ |
| Classifier learning rate | $5 \times 10^{-4}$ | Description weight $\lambda_{\text{desc}}$ | 0.01      |
| BM25 $k_1$ / $b$         | 1.2 / 0.75         | Retrieved abstracts per code               | 3         |
| Input length             | 4,000 tokens       | Early-stopping patience                    | 10        |
| Inference block size     | 256 codes          | Independent runs                           | 3         |

### S1.1. Facet-reference construction

The offline reference pipeline expands each canonical ICD description with UMLS synonyms and related concepts, then retrieves PubMed abstracts using BM25 [3, 4]. One request constructs the six facet records for a code so that their source context remains aligned. The cache key contains the dataset, taxonomy version, canonical description, and evidence snapshot. Each stored record contains the facet name, its reference description, source identifiers, and

ORCID(s):

a support flag. An unsupported facet remains in the cache with an empty description and a false support flag. The pipeline excludes patient notes, split assignments, test labels, and model predictions.

The following text reproduces the instruction supplied to the reference model.

**Instruction.** Produce evidence-grounded ICD reference descriptions using only the supplied canonical description, UMLS concepts, and PubMed abstracts. Exclude patient notes, coding labels, and unsupported claims.

**Inputs.** ICD code {code\_id}; canonical description {code\_description}; UMLS evidence {umls\_evidence}; retrieved abstracts {abstracts}.

**Output.** Provide one record for each of Definition, Symptoms, Diagnosis, Treatment, Complications, and Summary. Each record contains a facet name, a concise supported description, source identifiers, and a support flag. Leave the description empty when the supplied evidence does not support the facet.

Table S2 summarizes the full-taxonomy caches. The top-50 benchmarks reuse the corresponding records from their parent taxonomy rather than constructing a separate cache.

**Table S2**

Full-taxonomy data splits and cached reference coverage. PubMed coverage is the percentage of codes with retrieved support.

| Corpus          | Codes  | Train   | Valid. | Test   | Valid slots | Slots/code | PubMed (%) |
|-----------------|--------|---------|--------|--------|-------------|------------|------------|
| MIMIC-III ICD-9 | 8,922  | 47,723  | 1,631  | 3,372  | 41,672      | 4.67       | 91.4       |
| MIMIC-IV ICD-9  | 11,331 | 180,553 | 7,110  | 13,709 | 53,410      | 4.71       | 89.7       |
| MIMIC-IV ICD-10 | 26,096 | 110,442 | 4,017  | 7,851  | 119,230     | 4.57       | 86.2       |

## S2. Additional predictive metrics

Table S3 supplements the thresholded F1 results with ranking metrics. DescFuse has the highest Macro-AUC on the two MIMIC-III settings and the highest reported P@5 on both MIMIC-IV subsets. The result is not uniform across every metric: CL-LLM has higher P@5 on MIMIC-III-50, and MSMN and CoRelation have higher Micro-AUC on MIMIC-III Full. These differences separate code ranking from classification at validation-selected thresholds.

Table S4 reports run variation, area under the precision-recall curve (PR-AUC), and expected calibration error (ECE). We report PR-AUC as mean (SD) over three runs and calculate ECE from the evaluation probabilities under the validation-selected calibration protocol.

## S3. Sensitivity and objective controls

The one-factor sensitivity analysis varies retained note length, PubMed retrieval depth, and reference ordering (Table S5). Increasing the note or retrieval budget beyond the default changes Macro-F1 only slightly. Reducing either budget has a larger effect, especially on rare50. Reordering synonyms and permuting the facet instructions also lower the reported scores, while preserving the same training and evaluation procedure.

Table S6 consolidates the description-loss sweep, facet-set controls, and gradient-flow controls. The frozen variant attains the lowest description negative log-likelihood (NLL), but the joint objective yields the highest coding F1. Description fidelity alone therefore does not determine downstream utility.

## S4. Reference-source and evidence diagnostics

Table S7 varies the information supplied to reference construction while retaining the same model and evaluation procedure. The combined UMLS and PubMed condition has the highest valid-facet coverage and rare50 F1. PubMed alone yields higher coverage and description quality than UMLS alone but lower F1, showing that reference completeness and downstream utility are related but not interchangeable.

**Table S3**

Additional predictive metrics (%). Ma./Mi. denote Macro-/Micro-AUC. P@8 applies to MIMIC-III Full; P@5 applies to top-50 settings. Dashes denote unavailable results.

| Model           | MIMIC-III Full |             |             | MIMIC-III 50 |             |             | IV 9-50     | IV 10-50    |
|-----------------|----------------|-------------|-------------|--------------|-------------|-------------|-------------|-------------|
|                 | AUC-Ma.        | AUC-Mi.     | P@8         | AUC-Ma.      | AUC-Mi.     | P@5         | P@5         | P@5         |
| PLM-ICD         | 92.5           | 98.9        | 76.7        | 90.2         | 92.7        | 65.0        | 62.4        | 64.6        |
| KEPT            | –              | –           | –           | 92.6         | 94.8        | 67.3        | –           | –           |
| LAAT            | 91.9           | 98.8        | 73.8        | 92.5         | 94.6        | 67.5        | 62.0        | 64.4        |
| JointLAAT       | 92.1           | 98.8        | 73.5        | 92.5         | 94.6        | 67.1        | 62.0        | 64.5        |
| MSMN            | 95.0           | <b>99.2</b> | 75.2        | 92.8         | 94.7        | 68.0        | 62.6        | 65.2        |
| LCDL            | 95.2           | 98.9        | 75.9        | 93.1         | 95.1        | 67.8        | 62.4        | 64.8        |
| CoRelation      | 95.2           | <b>99.2</b> | 76.2        | 93.3         | 95.1        | 68.3        | 62.9        | 65.4        |
| LAAT+GPT        | 92.4           | 98.9        | 73.8        | 91.7         | 94.0        | 64.7        | 62.4        | 64.7        |
| CL-LLM          | 95.2           | 98.8        | 77.9        | 92.6         | 94.9        | <b>70.4</b> | –           | –           |
| BRSK            | 95.5           | 99.1        | 77.2        | 92.9         | 94.8        | 67.7        | 62.5        | 65.1        |
| <b>DescFuse</b> | <b>95.8</b>    | <b>99.1</b> | <b>78.0</b> | <b>93.7</b>  | <b>95.5</b> | <b>69.8</b> | <b>64.1</b> | <b>66.4</b> |

**Table S4**

Predictive performance and calibration of DescFuse. F1 values are percentages and reported as mean (SD) over three runs.

| Setting            | Macro-F1   | Micro-F1   | PR-AUC      | ECE   |
|--------------------|------------|------------|-------------|-------|
| MIMIC-III Full     | 11.9 (0.3) | 60.2 (0.4) | 0.72 (0.01) | 0.043 |
| MIMIC-III 50       | 70.5 (0.4) | 74.2 (0.3) | 0.79 (0.01) | 0.038 |
| MIMIC-III rare50   | 24.8 (0.3) | 25.8 (0.9) | 0.37 (0.02) | 0.055 |
| MIMIC-IV ICD-9-50  | 74.1 (0.3) | 77.0 (0.2) | 0.81 (0.01) | 0.045 |
| MIMIC-IV ICD-10-50 | 72.0 (0.4) | 76.0 (0.3) | 0.78 (0.01) | 0.048 |

**Table S5**

Input and reference sensitivity on MIMIC-III (F1, %).

| Factor          | Setting                | 50 Macro-F1 | rare50 Macro-F1 | rare50 Micro-F1 |
|-----------------|------------------------|-------------|-----------------|-----------------|
| Note length     | 2,000 tokens           | 69.8        | 24.1            | 25.1            |
| Note length     | 4,000 tokens (default) | 70.5        | 24.8            | 25.8            |
| Note length     | 6,000 tokens           | 70.7        | 25.0            | 25.9            |
| PubMed depth    | 1 abstract             | 69.7        | 23.8            | 25.0            |
| PubMed depth    | 3 abstracts (default)  | 70.5        | 24.8            | 25.8            |
| PubMed depth    | 5 abstracts            | 70.6        | 24.9            | 25.9            |
| Reference order | Synonyms reordered     | 70.2        | 24.6            | 25.5            |
| Reference order | Facets permuted        | 69.9        | 24.1            | 25.1            |

#### S4.1. Supported-facet strata

The note-level diagnostic uses only positive reference codes because negative codes have no ground-truth facet support. For note  $n$ , let  $C_n$  be its positive-code set and let  $f$  index Definition, Symptoms, Diagnosis, Treatment, Complications, and Summary. UMLS-expanded BM25 matches each facet reference to the sentences of the note using the same  $k_1 = 1.2$  and  $b = 0.75$  settings as the reference pipeline. The binary variable  $\text{support}(n, c, f)$  records whether each score meets the fixed matching criterion for code  $c$  and facet  $f$ . Note-level support takes the union across positive codes:

**Table S6**

Description-supervised training and architecture controls. Entries are three-run means; F1 values are percentages.

| Condition                                   | 50 Macro-F1 | rare50 Macro-F1 | Description NLL |
|---------------------------------------------|-------------|-----------------|-----------------|
| Joint model, $\lambda_{\text{desc}} = 0.01$ | <b>70.5</b> | <b>24.8</b>     | 1.48            |
| $\lambda_{\text{desc}} = 0$                 | 69.2        | 23.1            | 1.82            |
| $\lambda_{\text{desc}} = 0.001$             | 69.9        | 23.8            | 1.61            |
| $\lambda_{\text{desc}} = 0.1$               | 70.0        | 24.2            | 1.55            |
| One query                                   | 69.1        | 22.9            | 1.79            |
| Reduced facet set                           | 69.8        | 23.7            | 1.58            |
| Frozen after description pretraining        | 69.4        | 23.5            | <b>1.41</b>     |
| Stop-gradient to shared LoRA                | 69.8        | 23.9            | 1.57            |

**Table S7**

Reference-source controls on MIMIC-III rare50. Coverage is the percentage of valid facet records.

| Reference source           | Coverage (%) | Macro-F1    | Micro-F1    | Description score |
|----------------------------|--------------|-------------|-------------|-------------------|
| Canonical description only | 68           | 22.7        | 24.0        | 0.71              |
| Canonical + UMLS           | 77           | 24.1        | 25.4        | 0.78              |
| Canonical + PubMed         | 80           | 23.9        | 25.0        | 0.82              |
| Canonical + UMLS + PubMed  | <b>84</b>    | <b>24.8</b> | <b>25.8</b> | <b>0.85</b>       |

$$s_{n,f} = \max_{c \in C_n} \text{support}(n, c, f), \quad S_n = \sum_{f=1}^6 s_{n,f}. \quad (\text{S1})$$

The four strata are  $S_n = 0$ ,  $S_n \in \{1, 2\}$ ,  $S_n \in \{3, 4\}$ , and  $S_n \in \{5, 6\}$ . A note in the zero-support stratum can still have valid reference descriptions; it has no sentence match for any positive-code facet under this diagnostic.

Table S8 partitions all 3,372 MIMIC-III Full test notes. Notes and positive assignments are fixed data statistics. TP, FP, and FN are three-run mean counts. We calculate Micro-F1 from these mean counts. Within each run, we calculate Macro-F1 over all 8,922 codes and assign zero F1 to codes without positives in that stratum. The table reports the run mean, which cannot be reconstructed by weighting the four rows by note count.

**Table S8**

Performance by note-level supported-facet count on MIMIC-III Full. Null ECE values show mean (SD) over three runs.

| Facets | Notes | Pos.   | TP     | FP    | FN    | Micro-F1 | Macro-F1 all | Null ECE    |
|--------|-------|--------|--------|-------|-------|----------|--------------|-------------|
| 0      | 199   | 1,500  | 300    | 2,000 | 1,200 | 15.8     | 0.5          | 0.22 (0.02) |
| 1–2    | 653   | 6,000  | 2,000  | 4,000 | 4,000 | 33.3     | 2.5          | 0.18 (0.02) |
| 3–4    | 1,344 | 20,000 | 11,000 | 7,000 | 9,000 | 57.9     | 7.0          | 0.12 (0.01) |
| 5–6    | 1,176 | 26,115 | 18,921 | 8,238 | 7,194 | 71.0     | 16.0         | 0.09 (0.01) |

The summed mean counts give 60.18% pooled Micro-F1, which rounds to the 60.2% reported for MIMIC-III Full. The main result averages F1 across runs, so the agreement at the reported precision is a numerical consistency check rather than equality between aggregation rules.

## S4.2. Null-gate calibration

We evaluate null calibration over (note, positive code, facet) triples. The target  $y_i^\emptyset$  is the complement of the cached support flag after note-level matching, and the confidence  $p_i^\emptyset$  is the null-gate probability. We sort the triples into ten equal-frequency bins, skip empty bins, and do not merge adjacent bins. For nonempty bin  $\mathcal{B}_b$ ,

$$\text{conf}(\mathcal{B}_b) = \frac{1}{|\mathcal{B}_b|} \sum_{i \in \mathcal{B}_b} p_i^\emptyset, \quad (\text{S2})$$

$$\text{acc}(\mathcal{B}_b) = \frac{1}{|\mathcal{B}_b|} \sum_{i \in \mathcal{B}_b} y_i^\emptyset, \quad (\text{S3})$$

$$\text{ECE}_\emptyset = \sum_{b: |\mathcal{B}_b| > 0} \frac{|\mathcal{B}_b|}{N} |\text{acc}(\mathcal{B}_b) - \text{conf}(\mathcal{B}_b)|. \quad (\text{S4})$$

The support target comes from reference matching rather than a new clinician annotation. Null ECE therefore measures agreement with the matching-derived support labels.

## S5. Computational profile

Table S9 reports measurements collected on the same hardware. Trainable parameters exclude frozen weights. The profiler records peak memory with an inference block size of 256 codes. Separate runs provide latency and throughput, and all measurements exclude offline reference construction.

**Table S9**

Computational profile under the common profiling configuration.

| Model           | Trainable parameters (M) | Memory (GB) | Epoch (min) | Latency (ms/note) | Throughput (notes/min) |
|-----------------|--------------------------|-------------|-------------|-------------------|------------------------|
| PLM-ICD         | 125.4                    | 18.6        | 27.8        | 47                | 1,660                  |
| CoRelation      | 72.5                     | 16.3        | 22.1        | 44                | 1,780                  |
| BRSK            | 118.2                    | 17.1        | 24.7        | 38                | 2,060                  |
| <b>DescFuse</b> | <b>34.6</b>              | <b>15.5</b> | <b>19.4</b> | 41                | 1,920                  |

## S6. Representation-neighborhood statistics

The main manuscript visualizes matched code and evidence states for BRSK, the interaction-removal variant, and DescFuse. Table S10 reports the corresponding quantitative statistics in the normalized 24-dimensional export space before the two-dimensional PCA projection. Ten-neighbor purity excludes the query itself. Few-shot purity uses all 108 flagged codes, while the overall purity and silhouette scores use all 540 matched code identifiers.

**Table S10**

Clinical-chapter structure in the matched 24-dimensional representation exports.

| Representation  | Variant             | 10-NN purity | Few-shot purity | Silhouette   |
|-----------------|---------------------|--------------|-----------------|--------------|
| Code states     | BRSK                | 0.321        | 0.357           | 0.007        |
| Code states     | Without interaction | 0.746        | 0.725           | 0.090        |
| Code states     | DescFuse            | <b>0.917</b> | <b>0.899</b>    | <b>0.160</b> |
| Evidence states | BRSK                | 0.367        | 0.364           | 0.013        |
| Evidence states | Without interaction | 0.834        | 0.819           | 0.123        |
| Evidence states | DescFuse            | <b>0.945</b> | <b>0.946</b>    | <b>0.187</b> |

These statistics describe local chapter organization in the exported representations. They complement the coding and fusion controls but do not assign a causal role to the two-dimensional visualization.

## References

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